

Challenge 1 Kaleidoscope

```
#KaleidoscopeChallenge.py
```

```
import random
```

```
import turtle
```

```
t = turtle.Pen()
```

```
t.speed(0)
```

```
turtle.bgcolor('black')
```

```
colors=['red', 'yellow', 'blue', 'green',
```

```
        'orange', 'purple', 'white', 'gray']
```

```
for n in range(50):
```

```
    # generate spirals of random sizes/colors at random locations on the screen
```

```
    t.pencolor(random.choice(colors)) # pick a random color from colors[]
```

```
    size = random.randint(10,40)    # random size spiral from 10 to 40
```

```
sides = random.randint(3,9)    # random number of sides in spiral
```

```
thick = random.randint(1,6)    # random thickness of the lines
```

```
t.width(thick)
```

```
# generate a random (x,y) location on the screen
```

```
x = random.randrange(0,turtle.window_width()//2)
```

```
y = random.randrange(0,turtle.window_height()//2)
```

```
# first spiral
```

```
t.penup()
```

```
t.setpos(x,y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.left(360/sides + 2)
```

```
# second spiral
```

```
t.penup()
```

```
t.setpos(-x,y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.right(360/sides + 2)
```

```
# third spiral
```

```
t.penup()
```

```
t.setpos(-x,-y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
t.left(360/sides + 2)
```

```
# fourth spiral
```

```
t.penup()
```

```
t.setpos(x,-y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.right(360/sides + 2)
```